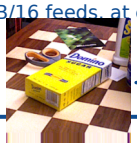
NeST: the network and the SE(3) decode are one top-down tree

a shared ViT-B/16 feeds, at every node, the block that computes that node's categorical; gold = the beam-MAP child, which becomes the next level's parent code



crop 224^2

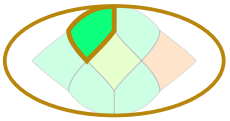
ViT-B/16 (augreg2)
12 transformer blocks
fine-tuned (lr 10^{-5})

patch tokens
$X \in \mathbb{R}^{196 \times 768}$

patch tokens X (K, V) condition every node

ROOT node — rotation ($\ell = 0$)

72 learned queries Emb(72×768)
cross-attn ×2 (Q; $K, V = X$)
72 logits → softmax



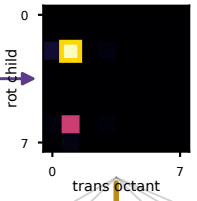
$p_0(c_0 \mid x)$
72-way rotation
(HEALPix S^2, hue = roll)

MAP child $p = 0.57$

chosen cell → next parent code

CHILD node ($\ell = 1$) — child_attn

parent code $[R, \hat{g}]$ ⊕ child offset
Fourier $\pi 2^k$ + child_proj MLP
cross-attn ×2 (Q; $K, V = X$)
64 logits → softmax



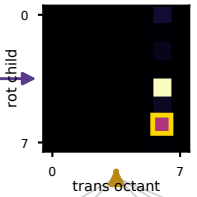
$p(c_1 \mid \cdot)$
64-way
(8 rot×8 oct)

MAP child $p = 0.28$

chosen cell → next parent code

CHILD node ($\ell = 2$) — child_attn (shared weights)

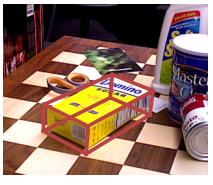
parent code $[R, \hat{g}]$ ⊕ child offset
Fourier $\pi 2^k$ + child_proj MLP
cross-attn ×2 (Q; $K, V = X$)
64 logits → softmax



$p(c_2 \mid \cdot)$
64-way
(8 rot×8 oct)

⋮ **child_attn reused, $\ell = 3, 4, 5$ (same block, deeper cells)**

leaf cell → $R =$ cell centre, $t = \hat{t} + Ag$



MAP 6D pose (R, t) → 3D box

every node is a log-softmax of a shared-weight attention block ⇒ the tree IS a normalized SE(3) density; zero-init output ⇒ starts at the uniform floor